MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4701: SOFTWARE ENGINEERING

END OF SEMESTER EXAMINATION

DATE: JULY 2023 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

There are **FOURTEEN** questions in this paper. **ANSWER ALL.**

There are **TWO** pages. **PLEASE CHECK**

Marks are indicated for each questions

1. State **three** components that encompass any software product. **[3 Marks]**
2. Explain any **three** major reasons that caused software crisis in the 1980s. **[6 Marks]**
3. Differentiate between the following:
    a. Prescriptive models and Agile models. **[2 Marks]**
    b. Functional Requirements and Non-functional Requirements **[2 Marks]**
4. Describe any **five** parameters you may use to assess or validate software requirements. **[10 Marks]**
5. Explain any **two** roles of refactoring in a software engineering process. **[4 Marks]**
6. Explain any **two** advantages that come with writing tests before writing the code. **[4 Marks]**
7. Describe any **three** challenges encountered during requirements gathering process. **[6 Marks]**
8. Discuss the Agile Manifesto. **[10 Marks]**
9. Explain any **three** examples of process model flow. **[6 Marks]**
10. Cloud Business Technologies is an e-commerce company that is planning to develop a market place platform. Marketplace connects buyers and sellers, facilitating online transactions. The system should support product browsing, shopping cart management, payment processing, and order fulfillment. Consider the system's requirements for handling a large number of products, facilitating secure transactions, and providing a seamless user experience. Assuming you are the Systems Engineer of the company:
    a. Suggest architectural style for this project. Justify your answer: **[4 marks]**
    b. Draw a high-level architecture diagram that illustrates the main components and their interactions based on the architecture style suggested in **10a** above. **[10 marks]**
    c. Describe any **three** ways on how the choice of architecture style affect the final system product. **[6 Marks]**
11. Cloud Business Technologies has hired you as the project manager for a CloudTunes music streaming platform software development project:
    a. Describe the process you would take to create and manage an effective project team, considering factors such as team composition, roles and responsibilities, and team dynamics. **[6 marks]**

b. Describe any **three** desirable traits this team must poses to accomplish project goals within allocated time and resources. **[6 Marks]**

c. Explain any **three** risks that can arise during development of this project. **[6 Marks]**

d. The CloudTunes project is facing budget constraints towards the end. Describe any **three** strategies you would employ to effectively manage the project budget, control costs, and ensure that the project is completed within the allocated financial resources. **[6 marks]**

12. A project size of 200 KLOC is to be developed. The software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the following:

a. The effort. **[2 Marks]**

b. Development time. **[2 Marks]**

c. Average staff size. **[2 Marks]**

d. Productivity of the project. **[2 Marks]**

13. Explain the role of community in an open source development environment. **[2 marks]**

14. Describe any **two** types of open source software licenses commonly used in the industry. **[4 marks]**

END OF QUESTION PAPER